Bryan Chen

https://yuyangchen23.github.io/personal-website/
https://github.com/yuyangchen23
bryanchen0923@gmail.com
+1 (604) 396-0795

B.Sc, Computing Science, Simon Fraser University 2021 - 2025

Technical Skills

- Programming Language: C++, Python, Kotlin, JavaScript, HTML, PostgreSQL, Rust
- Framework: Django, React
- Project management tools: Trello, Github
- Microsoft: Excel, PowerPoint, Word
- Operating System: Ubuntu, Mac OS, Windows

Professional Work Experience

Frontend Developer Contract Worker

Feb - May 2026

KTV Working Drone(https://www.ktvworkingdrone.com.tw) - Taichung, TW

- Developed a responsive frontend website using Next.js, React, TypeScript, and Tailwind CSS for KTV Working Drone's Taiwan business.
- Built dynamic service and news pages with reusable components, filtering, pagination, video embeds.
- Established Contentful Headless CMS and designed non-technical fields to enable co-workers to manage content easily.
- Implemented SEO foundations, including metadata, Open Graph tags, sitemap/robots support, canonical URLs, and structured JSON-LD for business, website, breadcrumbs, services, and news pages.

Frontend DeveloperIntern

May - Aug 2025

La Foret Julibee Cafe - Burnaby, BC

- Utilized modern-react tools to establish a fundamental layout for the restaurant's website
- Implemented designed system components to a specific project
- Optimized user interface, ensuring user-friendly UI across different devices

Transferable Skills

- Strong interpersonal and cross-team collaboration skills.
- Able to work in a high-paced environment.
- Able to communicate and receive information clearly via speaking and nonverbal clues.
- Capable of ensuring on-time completion of work and prioritizing activities efficiently.

Technical Project Experience

Implemented Multi-Agent path finding Algorithms

Dec 2025

Intelligent Systems (CMPT 417), SFU

- Developed and optimized Conflict-Based Search and Prioritized Planning algorithms in Python to coordinate collision-free paths for multiple agents in shared 2D environments.
- Architected a robust low-level search using A* in the space-time domain (x, y, t), integrating vertex and edge constraints to resolve complex spatial-temporal conflicts and swap-collisions between agents.
- Evaluated algorithm optimality and efficiency across industry-standard benchmarks (MovingAI maps, warehouses, mazes), utilizing Dijkstra-based heuristics and disjoint splitting techniques to minimize search-tree expansion.

2D Game Development Experience

Dec 2023

Introduction To Software Engineering (CMPT 276), SFU

- Designed and implemented a user interface for the game, ensuring smooth navigation.
- Gained a solid understanding of Java packages and OOP programming.
- Collaborated with peers in a team setting, adopting Agile methodologies to plan, develop, and utilize Git for version control.

Android App Development — Mar 2024

Mobile Applications Programming and Design (CMPT 362), SFU

- Implemented an Android app with multiple fragments and activities, featuring dynamic UI elements like dropdown menus, date pickers, and pop-up dialogs for user interactions.
- Integrated camera and gallery functionalities, ensuring smooth media handling with runtime permission management, enhancing the user experience for photo selection and capturing.
- Developed and customized Android PreferenceScreen for app settings, including radio buttons, checkboxes, and navigation links, improving configurability and user preference management.

Linux Memory Management Project — Jul 2025

Operating Systems I (CMPT 300), SFU

- Developed a custom shell with advanced command-line functionalities.
- Engineered and deployed a customized Linux kernel, successfully tested on a virtual machine environment.
- Implemented dynamic memory management techniques, including custom list structures and memory allocation strategies using 'malloc' and compaction methods.

Personal Project Experience

Personal Fincance App Project (private repository on github - ongoing project) — Nov 2025 - Mar 2026

- Designed as a reliable, mobile-friendly alternative to Google Sheets for portfolio tracking, specifically targeting users from the Canada community.
- Provided comprehensive account management (depository, credit, and investment), transaction tracking, and goal management, with built-in support for Canadian-specific financial features like TFSA, RRSP, and FHSA accounts.
- Built as an offline-first Progressive Web App (PWA) using a modern, type-safe stack that includes Next.js 15, tRPC, Prisma, PostgreSQL, and Supabase Auth, emphasizing a security-first architecture with comprehensive validation and audit logging.

Medical Platform Competition (https://github.com/yanjung/MediHacks2024) — Jul - Jul 2024

- Developed a user-centric and responsive web application, ensuring optimal performance.
- Integrated with third-party APIs to facilitate secure access and storage of patient data for future retrieval and analysis.
- Established robust connectivity between frontend and backend systems to ensure smooth and efficient functionality.

Personal Loan Project (https://github.sfu.ca/kha112/personal-loan-project) — Mar - Apr 2024

- Preprocessed and cleaned noisy data, employing various techniques to structure it into a well-organized dataframe.
- Conducted comprehensive statistical analyses on the organized data to identify variables with the highest correlation to loan borrowing.
- Implemented machine learning algorithms to predict an individual's eligibility and potential loan amount based on their financial status.